Data Networks Basics

Course objectives:

In today's interconnected world, understanding the fundamentals of data networks is essential. Whether you're an aspiring network engineer, an IT professional, or simply curious about how information flows across the digital landscape, this course will equip you with the foundational knowledge you need.

We'll dive into key concepts, starting with the bedrock of networking: **Ethernet** and the **Local Area Networks (LANs)**—the backbone of office networks, enabling seamless communication between computers, printers, and servers.

Extending the scope of networks, we'll move on into the wider world of **Wide Area Networks (WANs)**, where data crosses geographical boundaries, linking distant offices, data centers, and cloud services. Here we will meet the **Optical Transport Networks (OTNs)**, the foundation for the advanced MPLS and Metro-Ethernet networks. These fiber-optic marvels carry massive amounts of data across continents, making video streaming, cloud services, and telecommunication possible.

As technology evolves, so does our understanding of networking. Brace yourself for the paradigm shift brought by **Software-Defined Networking (SDN)**. Explore how SDN decouples control and data planes, enabling dynamic network management and agility.

Duration Instructional methods:

Option 1: On-Site Training: 3 days,
Classroom-style lecture using Power-point slides with interactive discussion, Live-network simulation presentation using Cisco Packet Tracer Software,
Questions & Answers after each module.

Option 2: Virtual Classroom: 3 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 6 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

The course is addressed to technical and non-technical staff from the technology, marketing and management field requiring an overview about the different data network solutions and architectures, standards, implementations and applications.

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Course Contents:

- Introduction to Computer Networks
- Data Network classification and topologies
- The OSI Model for Data Networks
- Devices for Data Networks: NIC's, Hubs, Switches, Routers, Gateways, Servers
- Addressing in Data Networks: MAC Address, IP Address, Port Numbers
- The Local Area Network
    - Introduction to Ethernet
    - Ethernet standards and the physical medium
    - The Hub-based LAN
    - Collision Domains and Broadcast Domains
    - Switch-based LAN
    - Virtual LAN - VLAN
    - Cisco Packet Tracer Simulation for LAN Architectures

- The Wireless LAN – WLAN
    - Introduction to WiFi
    - WiFi Network Architectures and Topologies
    - The WiFi Standards 802.11n, 802.11ac (WiFi 5), 802.11ad, 802.11ax (WiFi 6)

- Interconnecting LAN's – the basics of Routing
    - Routing with IPv4
    - Routing with IPv6
    - Some "Helper" protocols basics: ARP, DHCP and DNS
    - Cisco Packet Tracer Simulation for routing with IPv4

- The Metropolitan- / Wide-Area Network (MAN/WAN)
    - Network Classification: Access – Edge - Backbone
    - Optical Transmission Networks (OTN)
    - The basics of fiber-optical transmission in DWDM
    - The Multi-Protocol Label Switching (MPLS) Network
    - Carrier-Grade Ethernet and Metro-Ethernet

- Software-Defined Networking (SDN)
    - Introduction to Virtualization and SDN
    - The OpenFlow standard for SDN
    - Network Virtualization and VxLAN
    - SDN in Data Centers
    - Software-Defined WAN (SD-WAN) introduction

Pre-requisites:

Interest in Data Networks and IT.

Course Code: Data01